Introduction to Cryptography

Lecture 40: Hash Functions and MD5 Algorithm

Dr. Siddhant Thapliyal

Department of Computer Science and Engineering
Graphic Era (Deemed to be University), Dehradun

Learning Outcomes

- Define a cryptographic hash function.
- Explain preimage, second-preimage and collision resistance.
- Outline the MD5 processing stages.
- Compute and compare MD5 digests in Python.
- Explain why MD5 is unsuitable for security-critical uses.

What Is a Hash Function?

$$H : \{0,1\}^* \longrightarrow \{0,1\}^n$$

A variable-length input produces a fixed-length digest. The same input always produces the same digest.

Desired Hash Properties

- Preimage resistance: hard to find a message from its digest.
- Second-preimage resistance: hard to find a different message matching a given one.
- Collision resistance: hard to find any two different messages with equal digest.
- Efficient computation and strong avalanche behavior.

Hashing Is Not Encryption

Hashing	Encryption
No decryption key	Decryption possible with key
Fixed-length digest	Reversible ciphertext encoding
Used for integrity building blocks	Used for confidentiality

MD5 Overview

- Designed by Ronald Rivest.
- Processes data in 512-bit blocks.
- Produces a 128-bit (16-byte) digest.
- Uses four 32-bit state words and 64 main operations per block.
- MD5 collision resistance is broken; do not use it for digital signatures or collision-sensitive security.

MD5 Processing Stages

1. Append a 1 bit and enough zeros to reach 448 mod 512 bits.
2. Append the original message length as a 64-bit value.
3. Initialize four 32-bit state words.
4. Process each 512-bit block through four rounds (64 steps).
5. Concatenate final state words to obtain 128-bit digest.

MD5 Internal Functions

$$F(B, C, D) = (B \wedge C) \vee (\neg B \wedge D)$$

$$G(B, C, D) = (B \wedge D) \vee (C \wedge \neg D)$$

$$H(B, C, D) = B \oplus C \oplus D$$

$$I(B, C, D) = C \oplus (B \vee \neg D)$$

Each round mixes state, message words, constants, and rotations.

MD5 State

Four 32-bit chaining words: A, B, C, D.

$$\text{128-bit digest} = A \| B \| C \| D$$

Processing uses addition modulo 2^{32} and left rotations.

MD5 Demo: Python

Python 3

```
import hashlib
message = b"hello"
digest = hashlib.md5(message).hexdigest()
print(digest)
```

```
5d41402abc4b2a76b9719d911017c592
```

Avalanche Demonstration

Python 3

```
import hashlib
for message in (b"hello", b"Hello"):
 print(message, hashlib.md5(message).hexdigest())
```

A small input change causes a substantially different digest.

MD5 Security Limitations

- Practical collisions can be constructed.
- Therefore MD5 is not suitable for digital signatures, certificates, or adversarial integrity verification.
- A bare digest does not authenticate who created a message.
- Use SHA-256/SHA-3 for collision-resistant hashing; use HMAC for message authentication.
- For password storage, use a password hashing/KDF scheme such as Argon2id, not MD5 or raw SHA-256.

SHA-256 Comparison

Property	MD5	SHA-256
Digest size	128 bits	256 bits
Collision security	Broken	No practical general collisions known
Security use	Avoid	Commonly used

Demo Exercise

1. Hash two strings differing by one character.
2. Confirm MD5 outputs 32 hexadecimal characters.
3. Compute SHA-256 of the same strings.
4. Explain why equal MD5 digests do not prove equal messages.

Summary

MD5: 512-bit blocks $\longrightarrow$ 128-bit digest

Understand MD5 historically; choose modern hash functions for security.

Your Feedback Matters!

Please scan the QR code and share your feedback.



TCS-392 – Introduction to Cryptography